Perception-Constrained Altitude Control for Aerial–Ground Cooperative Tracking with a Shared Downward Camera

***Abstract*— We present a reproducible software-in-the-loop (SITL) study of aerial–ground cooperative tracking in which a single downward UAV camera is the ground vehicle's sole exteroceptive sensor, serving simultaneously as its navigation reference (path segmentation) and its state estimator (ArUco/EKF). This shared-sensor architecture couples navigation and estimation through a single control variable - the UAV altitude - and this coupling is the organizing principle of our design: low altitude sharpens estimation (ground-sampling noise $R \propto z^2$) but starves navigation coverage (the path lookahead leaves the camera footprint), while high altitude does the reverse. We develop a perception-constrained altitude policy guided by the shared-camera coverage constraint and demonstrate it closed-loop in SITL.**

I. INTRODUCTION

The deployment of Unmanned Ground Vehicles (UGVs) in unstructured or GPS-denied environments is heavily constrained by their limited perception horizons. Onboard sensors such as LiDAR and stereo cameras have been used in such vehicles for navigation purposes, but they suffer from occlusion, narrow fields of view (FoV), and an inability to anticipate the upcoming path. To mitigate this, aerial–ground cooperative systems employ anUnmanned Aerial Vehicle (UAV) as an elevated "eye in the sky" for a perception-limited UGV.

Most cooperative-tracking frameworks treat the UAV as a passive observer: it follows a predefined trajectory or simply minimizes distance to the UGV, treating the UGV's state estimation as a downstream byproduct [3]. This decoupling is fragile: when the UGV maneuvers aggressively or is occluded, the UAV can lose the target, the estimation covariance grows, and tracking degrades. Active-perception planning, where the sensor trajectory is optimized to maximize information gain [4], addresses part of this but typically relies on continuous, computationally expensive replanning and rarely integrates robust tracking guarantees for the aerial vehicle.

We present a shared-camera perception-control architecture in which both vehicles adapt to the ground vehicle's estimation uncertainty. Its contributions, as realized in the accompanying open implementation, are:

- A perception-constrained altitude policy, formalized and demonstrated closed-loop. With one downward camera serving both the ground vehicle's navigation (path segmentation) and its state estimate (ArUco/EKF) [5], [6], low altitude sharpens estimation ($R \propto z^2$) but starves navigation coverage; the policy flies as low as segmentation allows and climbs into turns.
- We show a perception-driven event trigger cuts the UAV nominal (aerial) planner's online re-solves $\approx 25\times$ ($797 \pm 16 \rightarrow 32 \pm 1$ per run) and its CPU $\approx 12\times$ at matched estimation quality (covariance 0.0095, $\approx 99.9\%$ visibility for both continuous and event-triggered planning).
- A reproducible SITL realization in ROS 2/Gazebo (vision/ArUco pipeline, view-quality-coupled EKF, belief-space UGV MPC, and the UAV planner/ancillary pair [1], [7]). Code and data are openly available.[1]

The remainder of the paper is organized as follows. Section II reviews related work; Section III states the problem; Section IV gives the vehicle and perception models; Section V develops the coupled event-triggered MPC; Section VI states the theoretical properties and the exact implementation status; and Section VII reports the simulation study.

II. RELATED WORK

Tube MPC [1], [8], [10] decomposes control into a nominal plan and an ancillary law that confines the true state to a robust positively invariant (RPI) tube; Raković et al. [2] give the standard outer approximation of the minimal RPI set, and the nonlinear case is treated in [9]. These guarantees assume a fixed, bounded disturbance set and constraint tightening computed offline. Time-varying (homothetic/parameterized) tubes generalize the fixed tube but require the scaling to enter the optimization with its own feasibility argument; our online adaptivity (Section V) is a lighter-weight surrogate.

Perception-aware trajectory optimization for quadrotors [4], [14] and active visual localization [15] shape motion to serve estimation. Belief-space planning [5], [6], [16], [18] augments the planning state with the estimator covariance helping planning under partial observability [17] and of active SLAM [20], [19]. Our belief-space UGV controller is in this family. Our architecture is bidirectional: the target shapes its trajectory to stay observable while the tracker adapts both its altitude and its robustness margin to perception quality.

Cooperative aerial–ground teams. Using an aerial vehicle as an "eye in the sky" for a ground robot has been demonstrated [23], including recent perception-limited-UGV navigation [21] and aerial-vision guidance of a perception-less ground vehicle [22], alongside quadrotor target-tracking under field-of-view and occlusion constraints [24], [25] and

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[1] https://github.com/shlok-mehndiratta/ugv_uav_

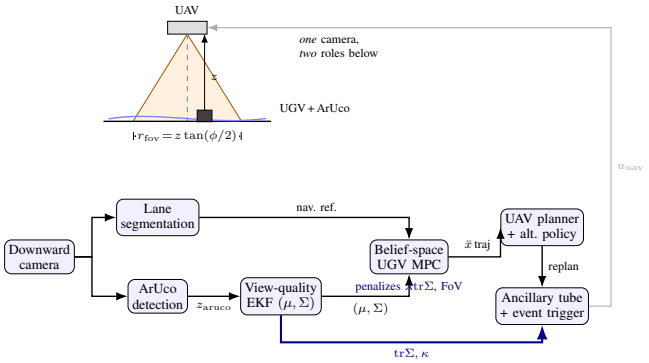


Fig. 1. System overview. Top: the physical geometry — a downward-camera UAV at altitude z views an ArUco-tagged UGV on a curving lane within a footprint of radius $r_{\text{fov}} = z\tan(\phi/2)$. Bottom: the single camera image feeds two roles: lane segmentation provides the UGV's navigation reference, and ArUco detection feeds the view-quality EKF (μ, Σ); both drive the belief-space UGV MPC (which penalizes $\lambda\,\text{tr}\Sigma$ and FoV exit), whose predicted trajectory the UAV planner tracks under the perception-constrained altitude policy, executed by the ancillary tube/event-trigger controller (blue: the EKF covariance and curvature drive the event trigger). This dual role of one sensor is the coverage–resolution coupling studied in the paper.

EKF-based cooperative localization [26], [27]. These systems typically fix the aerial vehicle's altitude and treat it as a passive relay; we show instead that altitude is an active perception control input whose optimal use is dictated by the shared-camera coupling.

vent-triggered and self-triggered control [12], [11] re-solve only when a state-dependent condition fires, and extend to robust MPC of nonlinear systems [7], [13]. We instantiate a perception-driven trigger with a critical tube-exit condition plus anticipatory conditions.

III. PROBLEM FORMULATION

A UAV tracks a UGV in a global frame $\mathcal{F}_W$. The UAV carries the only exteroceptive sensor: a downward monocular camera that detects an ArUco marker on the UGV, yielding a noisy pose measurement whose quality depends on the imaging geometry. An EKF fuses these detections with UGV wheel odometry to produce a belief (μ, Σ) over the UGV state. The objective is to (i) keep the UGV inside the camera footprint so that Σ stays bounded, (ii) track the UGV robustly under bounded disturbances, and (iii) do so while limiting the number of online optimizations. The UGV must track a ground lane while remaining observable.

IV. SYSTEM ARCHITECTURE AND MODELS

Fig. 1 shows the implemented system and its signal flow.

A. UGV dynamics

The UGV uses unicycle kinematics with first-order actuator lag. Let $\mathbf{x}_g = [x_g, y_g, \theta, v, \omega]^T$ be the state, (x_g, y_g) the position, θ the heading, (v, ω) the linear/angular velocities, and $\mathbf{u}_g = [v_c, \omega_c]^T$ the commanded velocities. The discrete-time dynamics are

$$\mathbf{x}_{g,k+1} = \mathbf{f}_g(\mathbf{x}_{g,k}, \mathbf{u}_{g,k}) + \mathbf{w}_g, \tag{1}$$

where the actuator lag is governed by time constants τ_v, τ_ω and $\mathbf{w}_g$ is process noise.

B. UAV model

Although the platform is a quadrotor, the planner and the ancillary tracker treat it as a velocity-controlled point mass (single integrator); an inner-loop attitude/thrust controller, provided by the Gazebo velocity-control plugin, is assumed to realize commanded velocities. With position $\mathbf{p} = [p_x, p_y, p_z]^T$ and velocity command $\mathbf{u} = [v_x, v_y, v_z]^T$,

$$\mathbf{p}_{k+1} = \mathbf{p}_k + \Delta t\,\mathbf{u}_k + \mathbf{w}_k, \qquad A_d = I_3,\ B_d = \Delta t\, I_3, \tag{2}$$

with bounded disturbance $\mathbf{w}_k \in \mathbb{W}$. All three axes are fully actuated: v_z is a genuine decision variable and the altitude policy of Sec. IV-D is realized closed-loop in SITL (the UAV altitude spans $[2.5, 7]$ m; Sec. VII).

C. Ground-sampling-distance measurement model

The accuracy of visual localization depends on the camera's ground sampling distance (GSD), which increases linearly with UAV altitude. Assuming the image-space localization error remains approximately constant in pixels, the corresponding metric position error grows proportionally to the GSD, implying that the position-measurement covariance scales quadratically with altitude. Let z denote the UAV altitude, z_0 a reference altitude, $d = \|\mathbf{p}^{xy} - \mathbf{p}_g\|$ the horizontal UAV–UGV separation, and $r_{\text{fov}}(z) = z\tan(\phi/2)$ the camera footprint radius. The measurement covariance is modeled as

$$R(z) = R_0 \left(\frac{z}{z_0}\right)^2 \left(1 + \left(\frac{d}{r_{\text{fov}}}\right)^2\right), \tag{3}$$

where R_0 is the baseline ArUco measurement covariance at the reference altitude z_0. The second term provides a mild inflation of the covariance as the target approaches the edge of the camera footprint, accounting for the degradation in localization accuracy away from the image center.

Equation (3) captures the resolution side of the coverage–resolution tradeoff: increasing altitude enlarges the camera footprint and improves navigation coverage (Sec. IV-D), but reduces image resolution, causing the measurement covariance to grow as $R \propto z^2$. This GSD-based model is used by both the EKF and the belief-space planner in all reported experiments, ensuring that planning propagates the same measurement uncertainty as the estimator.

D. Segmentation coverage model

The same downward camera also supplies the UGV's navigation reference: an OpenCV segmenter extracts the lane and seeds a waypoint a lookahead distance ahead of the vehicle. That waypoint must lie inside the camera footprint or segmentation fails and the UGV loses its reference. With footprint radius $r_{\text{fov}}(z) = z\tan(\phi/2)$ and required lookahead $d_{\text{look}}(\kappa)$ — which grows with local path curvature κ, since a curving lane exits the footprint sooner and needs more forward view — define the non-dimensional coverage factor

$$\eta = \frac{r_{\text{fov}}(z)}{d_{\text{look}}(\kappa)} = \frac{z\tan(\phi/2)}{d_{\text{look}}(\kappa)}, \tag{4}$$

with segmentation feasible for $\eta \gtrsim 1$. This is the crux coupling: the estimator wants low altitude ($R \propto z^2$, better resolution), while (4) demands sufficient altitude for navigation. The estimation-optimal altitude is thus the lowest that keeps $\eta \gtrsim 1$, and this floor rises with curvature — the analytic basis for climbing into turns. The aerial planner (Sec. V-B) enforces $\eta \gtrsim 1$ as a soft lower bound; Sec. VII reports the empirical cost tradeoff over η.

V. Coupled Event-Triggered Adaptive Control

The estimation uncertainty Σ_k enters the control formulations of both vehicles.

A. Information-aware UGV control (belief-space NMPC)

The UGV controller augments the planning state with the covariance, $\xi_k = (\mu_k, \Sigma_k)$, and propagates both through the horizon. At each predicted step it forms the no-observation prediction $\Sigma_{\text{pred}} = F\Sigma F^T + Q$ and a post-measurement covariance Σ_{obs} (Joseph update), and blends them by a smooth visibility gate $\gamma = 1/(1 + e^{\beta(d - r_{\text{fov}})})$:

$$\Sigma_k = (1-\gamma)\,\Sigma_{\text{pred}} + \gamma\,\Sigma_{\text{obs}}. \tag{5}$$

The cost minimizes tracking error while penalizing predicted uncertainty and FoV exit,

$$J_g = \sum_{k=0}^{N_g-1} \Big(\|\mathbf{e}_{g,k}\|_Q^2 + \|\mathbf{u}_{g,k}\|_R^2 + \lambda\ \mathrm{tr}(\Sigma_k) + w_{\text{fov}}\, c_{\text{fov}}(\mathbf{x}_{g,k}) \Big), \tag{6}$$

where the FoV term is a one-sided quadratic hinge $c_{\text{fov}} = \big(\max(0,\, d_k - r_{\text{fov}})\big)^2$ (a soft constraint, not an asymptotic barrier). Crucially, the belief rollout (5) uses the same measurement-noise model as the on-board EKF so the planner propagates the same uncertainty model used by the EKF. The covariance rollout (5) makes (6) a nonconvex NLP, solved with SLSQP over $N_g = 5$. The two terms $\lambda\,\mathrm{tr}(\Sigma_k)$ and $w_{\text{fov}} c_{\text{fov}}$ are what make the controller belief-coupled and are absent from the passive baseline.

B. Aerial nominal planner and altitude policy

The UAV computes a nominal trajectory $\bar{\mathbf{p}}$ over $N_a = 10$ by optimizing goal tracking, an altitude regularizer toward z_{opt}, and a visibility term that tolerates lateral offset only when altitude is sufficient:

$$J_a = \sum_k w_{\ell,k} \Big[\|\bar{\mathbf{p}}_k - \mathbf{g}\|^2 + w_z (z_k - z_{\text{opt}})^2 + w_{\text{alt}} \frac{\|\bar{\mathbf{p}}_k^{xy} - \mathbf{g}^{xy}\|^2}{z_k^2 + 0.1} \Big] + w_u \|\bar{\mathbf{u}}_k\|^2, \tag{7}$$

where $\mathbf{g}$ is the UGV's predicted end-of-horizon position (read from the belief-MPC rollout, one direction of the coupling) and $w_{\ell,k}$ up-weights later steps. The perception-constrained altitude policy of Sec. IV-D sets the altitude setpoint explicitly to the coverage floor plus an anticipatory margin,

$$z_k \;=\; \hat{d}_k / \tan(\phi/2) \;+\; \beta_\kappa\, \hat{\kappa}_k \;+\; z_{\text{m}}, \tag{8}$$

where $\hat{d}_k$ is the estimated horizontal UAV–UGV offset (the first term keeps the target within the footprint), $\hat{\kappa}_k$ is the anticipated path curvature, and $\beta_\kappa \hat{\kappa}_k$ raises altitude into turns where the required lookahead $d_{\text{look}}(\kappa)$ — and hence the coverage floor $\eta \gtrsim 1$ (4) — grows; z_{m} is a fixed margin that keeps the forward lookahead in view on straights. "Climb into turns to widen the footprint" is thus a derived consequence of the coverage constraint, not a hand-coded rule.

C. Offline robust-invariant-set baseline

The ancillary law is $\mathbf{u}_k = \bar{\mathbf{u}}_k - K\mathbf{e}_k$ with tracking error $\mathbf{e}_k = \mathbf{p}_k - \bar{\mathbf{p}}_k$; substituting into (2),

$$\mathbf{e}_{k+1} = \underbrace{(A_d - B_d K)}_{=:A_K} \mathbf{e}_k + \mathbf{w}_k, \qquad \mathbf{w}_k \in \mathbb{W}. \tag{9}$$

For A_K Schur and $\mathbb{W}$ compact, the minimal RPI set is outer-approximated offline by a truncated Minkowski series and the constraints tightened by the Pontryagin difference [2],

$$\Omega = \bigoplus_{i=0}^{N} A_K^i \mathbb{W} \supseteq \Omega_\infty, \quad \bar{\mathbb{X}} = \mathbb{X} \ominus \Omega, \quad \bar{\mathbb{U}} = \mathbb{U} \ominus K\Omega. \tag{10}$$

This serves as a design-time baseline: Ω is a certified robust bound for the idealized system and seeds the ellipsoidal shape matrix P_0 below.

D. Online adaptive ellipsoidal tube

Propagating the polytope Ω online — vertex/support-function updates and a Pontryagin difference at every 20 Hz ancillary step — is impractical on the UAV's embedded companion processor. Rather than pay that overhead, the ancillary controller uses an ellipsoidal invariant-set surrogate

$$\mathcal{E}(\alpha_k) = \{\mathbf{e} : \mathbf{e}^T P_0\, \mathbf{e} \le \alpha_k^2\}, \tag{11}$$

with an isotropic shape matrix $P_0 = 0.186\, I_3$ and a radius that adapts to the EKF horizontal covariance trace $\mathrm{tr}\,\Sigma_{xy} = \Sigma_{xx} + \Sigma_{yy}$:

$$\alpha_k = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})\,\mathrm{clip}\Big(\tfrac{\mathrm{tr}\,\Sigma_{xy} - \mathrm{tr}_{\min}}{\mathrm{tr}_{\max} - \mathrm{tr}_{\min}}, 0, 1\Big), \tag{12}$$

($\alpha_{\min} = 0.5,\ \alpha_{\max} = 2$). The tube breathes: it tightens when the estimate is sharp and loosens when perception degrades. Here α_k is the lightweight run-time monitor/trigger radius of the offline-synthesized tube; the certified objects remain the offline mRPI set Ω (10) and the A_K-Schur ISS bound (§VI), so tuning $\alpha_{\min}$ sets trigger sensitivity, not invariance. A full homothetic tube that re-derives (10) online with a matching recursive-feasibility argument is a natural extension.

E. Perception-driven event trigger

The ancillary node runs at 20 Hz but requests a nominal replan only when one of four prioritized conditions fires (implemented literally as follows).

- Tube exit (critical, no lockout): fire iff $\mathbf{e}_k^T P_0 \mathbf{e}_k > \alpha_k^2$, i.e. the state left the adaptive ellipsoid (11).
- Covariance spike: $\mathrm{tr}\,\Sigma_{xy}(t) > \tau_{\text{cov}}$ and a sudden ($> 1.5\times$) rise over ~ 0.5 s (e.g. ArUco occlusion).

- Curvature anticipation: the discrete curvature of the UGV predicted path, $\hat{\kappa} = \max_i |\Delta\theta_i|/\Delta s_i$ — the wrapped heading change $\Delta\theta_i$ between consecutive predicted waypoints over their spacing Δs_i, maximized over the next few steps — exceeds $\tau_\kappa = 0.4$ rad/m, so the UAV climbs before the turn (raising the coverage floor of Sec. IV-D).
- Temporal drift: elevated $\mathrm{tr}\,\Sigma_{xy}$ with time since last replan exceeding $T_{\min}$.

Conditions (2)–(4) carry a 1 s time-lockout (the chattering guard); condition (1) has none. Between triggers the ancillary law regulates the error via (9), and the previous nominal sequence is shifted forward.

F. Nominal optimization actually solved

At each trigger the planner solves (7) over the single integrator (2) with velocity bounds $v_x, v_y \in [-1.5, 1.5]$, $v_z \in [-1, 1]$ m/s, warm-started from the previous solution (SLSQP, maxiter $= 15$). Altitude limits $z \in [z_{\min}, z_{\max}]$ are shaped by the cost; a hard terminal set $\mathbb{X}_f$ and the DARE terminal cost of the idealized formulation below are not enforced online.

VI. From Offline Robust Design to the Online Controller

The implemented controller follows a two-stage design philosophy. First, an offline robust-MPC synthesis provides the ancillary feedback gain, invariant-set geometry, and constraint-tightening framework. Second, the online controller replaces the computationally intensive polyhedral tube updates with the adaptive ellipsoidal surrogate of Sec. V, enabling real-time execution on a compute limited companion computer.

Offline robust design. For the error dynamics (9), with A_K Schur and a bounded disturbance set $\mathbb{W}$, a conventional tube-MPC formulation computes a robust positively invariant (RPI) set Ω together with tightened state and input constraints $\bar{\mathbb{X}}, \bar{\mathbb{U}}$ [1], [2]. These offline objects provide the reference tube and determine the shape matrix P_0 used by the online ellipsoidal approximation.

Online implementation. Rather than propagating the full polyhedral tube online, the implemented controller employs the adaptive ellipsoidal surrogate (11), whose radius varies according to the EKF covariance. This approximation substantially reduces online computation while preserving a lightweight mechanism for monitoring tracking uncertainty. Between replanning events, the ancillary controller regulates the tracking error according to (9); for the implemented gain $K = \mathrm{diag}(0.6, 0.6, 0.8)$ at the Δt=0.05 s step (discrete error map $A_K = I - \Delta t\, K = \mathrm{diag}(0.97, 0.97, 0.96)$, Schur), bounded disturbances imply bounded inter-trigger tracking error, making ellipsoid membership a practical triggering metric.

Unlike a classical tube-MPC implementation, the online controller does not explicitly enforce the terminal ingredients required for recursive-feasibility guarantees. Consequently, the proposed controller should be viewed as a computationally efficient approximation inspired by the offline robust design, whose closed-loop properties are evaluated empirically in Sec. VII.

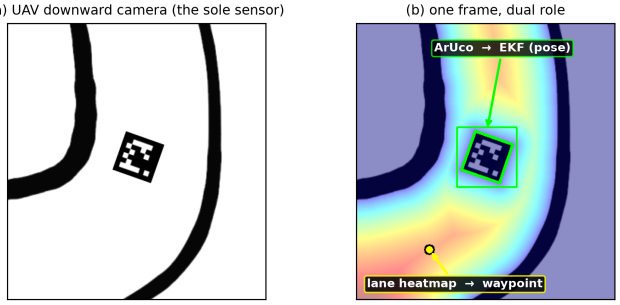


Fig. 2. The single downward camera in operation (real frame from the headless SITL system). (a) A raw UAV camera image: the curved lane with the UGV's ArUco marker. (b) The same frame drives both roles — the segmented-lane distance-transform heatmap yields the navigation waypoint (yellow), while the detected ArUco (green) feeds the EKF pose. This is the shared-sensor coupling the paper studies: one image, two jobs.

Scope and design insight. The experimental evaluation uses a velocity-controlled UAV abstraction appropriate for the high-level planning layer. The central architectural feature is that a single downward camera simultaneously provides the UGV's navigation reference and state estimate. This shared sensing creates an inherent coverage–resolution tradeoff governed by UAV altitude, motivating the perception-constrained altitude policy developed in this work. Extending the framework to a full multicopter model and hardware experiments remains future work.

VII. Simulation Study

Setup. The system runs in Gazebo with ROS 2 (Fig. 2): a TurtleBot3-class UGV with an ArUco marker, an X500-class UAV with a downward camera, an OpenCV lane/marker pipeline, the 5-state EKF, the belief-space UGV MPC, and the UAV planner/ancillary pair. Per-step telemetry (pose, ground truth, $\mathrm{tr}\,\Sigma$, FoV distance, ArUco visibility, view quality, replan events, solve times) is logged by a common logger identical across configurations.

Co-design: event-triggering makes the aerial planner affordable. The belief-space UGV controller runs continuously — it is the ground vehicle's feedback loop — so the compute lever is the UAV nominal (aerial) planner that re-optimizes the aerial tracking trajectory. Continuous aerial replanning (C1) spends a large, sustained CPU budget; the perception-driven event trigger (C2) re-solves it only when a covariance/curvature/drift condition fires. On the nominal track (headless, R=3), C2 reduces aerial-planner invocations $\approx 25\times$ ($797 \pm 16 \to 32 \pm 1$) and its CPU $\approx 12\times$ ($3123 \pm 56 \to 253 \pm 6$ ms per 45 s run; Table I), at matched estimation quality (steady-state $\mathrm{tr}\,\Sigma = 0.0095$ and $\approx 99.9\%$ ArUco visibility for both). Per-solve time is slightly higher for C2 (7.9 ± 0.4 vs. 3.9 ± 0.1 ms, a warm-start effect), but the $\approx 25\times$ reduction in solve count contributes towards net computational advantages. This is the co-design point: event-triggering makes perception-aware aerial replanning

affordable on a constrained companion computer at estimation quality indistinguishable from continuous replanning. The one cost is a small loss of inter-trigger smoothness — applying the previous nominal between triggers raises high-frequency trajectory content — a favorable tradeoff. Isolating how much the perception-specific trigger conditions add beyond a plain error-triggered baseline is a focused ablation for follow-on work.

TABLE I

AERIAL-PLANNER CO-DESIGN (HEADLESS, TRACK2, MEAN $\pm$ STD, R=3). C1 CONTINUOUS VS. C2 EVENT-TRIGGERED; THE BELIEF-SPACE UGV CONTROLLER RUNS CONTINUOUSLY IN BOTH. STEADY-STATE $t \geq 8$ S.

Metric	C1 (cont.)	C2 (event)
Aerial solves (#/run)	797 ± 16	32 ± 1
Mean solve (ms)	3.9 ± 0.1	7.9 ± 0.4
Aerial CPU (ms/run)	3123 ± 56	253 ± 6
Steady-state $\mathrm{tr}\,\Sigma$	0.0095	0.0095
ArUco vis. (%)	99.8	99.9
UAV–UGV offset (m)	0.128 ± 0.002	0.060 ± 0.005

The coverage–resolution altitude tension. With the GSD model enabled, we sweep altitude across three tracks of increasing curvature (headless; low-altitude cells are the median of R=3 reps). Two costs move oppositely with altitude (Fig. 3): estimation uncertainty $\mathrm{tr}\,\Sigma$ decreases as altitude decreases (0.0094 at z=6 m to 0.0053 at z=2 m — the $R \propto z^2$ resolution gain, and track-independent), while the UGV steering effort needed to keep the path segmented increases as altitude decreases, and increases with path curvature. This is the tension the altitude policy must resolve: fly as low as segmentation allows, and climb where the navigation floor rises.

We close the loop with the perception-constrained policy and compare it against the two natural fixed-altitude baselines on the hairpin (the most demanding track; headless, GSD noise, R=3). The policy actuates altitude over $[2.5, 5.1]$ m — holding near the low-altitude floor on straights and climbing through turns (Fig. 4) — and it is not matched by any static choice (Fig. 4). The fixed-low baseline (z=2.5 m) obtains good average resolution but suffers coverage-loss transients: as the target leaves the small footprint in the hairpin, its peak covariance spikes to 0.14 ($9\times$ the adaptive policy's peak of 0.016) and it drops marker visibility. The fixed-high baseline (z=7 m) never loses coverage but pays a 61% estimation penalty ($\mathrm{tr}\,\Sigma = 0.0105$ vs. 0.0065) from the degraded GSD. The adaptive policy attains the low mean covariance of flying low (0.0065, matching the fixed-low 0.0062) while keeping the peak covariance and FoV exits of flying high, demonstrating a favorable balance between estimation accuracy and coverage.

Consistent with the shared-camera architecture, the estimation uncertainty $\mathrm{tr}\,\Sigma$ at a given altitude is largely determined by the measurement noise model and is therefore nearly track-independent. In contrast, the navigation cost depends strongly on path geometry, so altitude must balance estimation quality against sufficient camera coverage. At the most demanding operating point (z = 2 m, UGV speed 1.0 m/s), occasional simulator timing variations lead to small run-to-run differences in marker visibility. Rather than identifying a sharp feasibility boundary, we therefore characterize the tradeoff through the overall performance trends and the comparison between adaptive and fixed-altitude policies.

TABLE II

SYSTEM PARAMETERS AND CONTROLLER HYPERPARAMETERS*

Subsystem	Parameter	Value
Camera / EKF	Camera FoV (ϕ)	1.047 rad
	Baseline Altitude (z_0)	3.0 m
	Baseline Covariance (R_0)	2.5×10^{-3} m^2
	EKF Update Rate	20 Hz
UGV NMPC	Horizon (N), Step (dt)	5, 0.2 s
	Cost Weights ($Q/R/w_{fov}$)	10 / 10 / 5
	Covariance Penalty (λ)	1.0
	Bounds (v_{max}, ω_{max})	0.22 m/s, 2.0 rad/s
UAV Planner	Horizon (N), Step (dt)	10, 0.2 s
	Velocity Bounds (v_x, v_y, v_z)	$\pm 1.5, \pm 1.5, \pm 1.0$ m/s
	Altitude Weight (w_{alt})	50
	Policy (z_m, β_κ)	0.4 m, 5.0
Tube & Triggers	Shape Matrix (P_0)	$0.186 I_3$
	Adaptive Tube Radius (α)	$[0.5, 2.0]$
	Cov. Triggers (spike, drift)	0.12, 0.07
	Timing (T_{min}, lockout)	3.0 s, 1.0 s

* **Note on Experimental Configurations:** The co-design validation (C1/C2) utilized a fixed reference altitude of $z_{opt} = 6.0$ m to isolate computational metrics. The adaptive altitude policy experiments dynamically actuated altitude within the bounded operational band of $[2.5, 7.0]$ m.

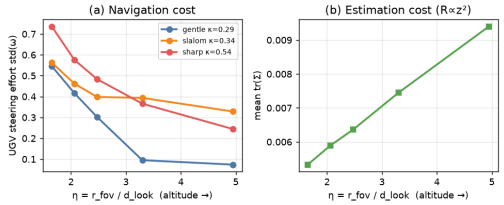


Fig. 3. The two jaws of the altitude tension ($\eta = r_{\mathrm{fov}}/d_{\mathrm{look}}$, increasing with altitude). Left: UGV steering effort (navigation cost) rises at low altitude and with path curvature. Right: estimation uncertainty $\mathrm{tr}\,\Sigma$ falls at low altitude (GSD, $R \propto z^2$; track-independent). The optimum flies as low as segmentation allows.

VIII. CONCLUSIONS

We studied aerial–ground cooperative tracking where a single downward UAV camera is the ground vehicle's sole exteroceptive sensor, serving both its navigation reference and its state estimate. We identified the coverage–resolution altitude tension intrinsic to this shared-sensor architecture and demonstrated closed-loop that the perception-constrained altitude policy resolving it achieves lower peak estimation error than the low-altitude baseline ($9\times$ smaller peak covariance) while avoiding the sustained estimation penalty of the high-altitude baseline (61% estimation penalty)

We further showed a perception-driven event trigger cuts the UAV nominal (aerial) planner's re-solves $\approx 25\times$ and its CPU $\approx 12\times$ at matched estimation quality, and we release the reproducible testbed. Future work: a full multicopter

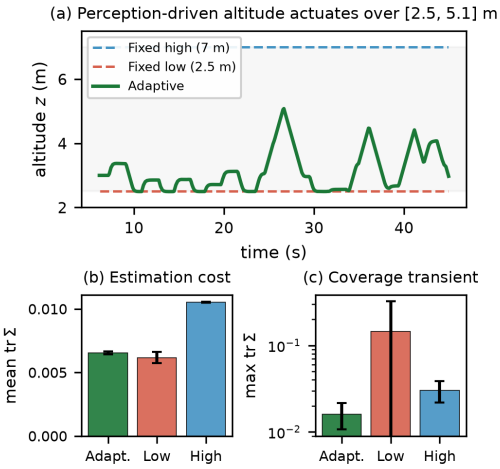


Fig. 4. Adaptive altitude outperforms fixed altitude (hairpin, headless, GSD noise, R=3). (a) The policy actuates altitude over $[2.5, 5.1]$ m, holding near the low-altitude floor on straights and climbing through turns, versus the two static baselines. (b) A fixed-high policy pays a 61% mean-estimation penalty. (c) A fixed-low policy suffers coverage-loss transients ($9\times$ larger peak $\mathrm{tr}\,\Sigma$, log scale). The adaptive policy achieves a favorable balance between estimation accuracy and coverage.

model for hardware-faithful altitude dynamics; a second exteroceptive sensor to isolate the observability benefit; and a homothetic tube carrying recursive-feasibility guarantees into the online loop.

A. Limitations and Future Work

The present study is limited to software-in-the-loop evaluation using a velocity-controlled UAV abstraction and a single-camera sensing architecture. While the proposed controller is motivated by an offline robust-MPC design, the implemented online controller employs a computationally efficient ellipsoidal surrogate rather than a full tube-MPC formulation. Consequently, the results should be interpreted as an empirical evaluation of the proposed perception-aware architecture under controlled simulation conditions.

Future work will extend the framework to hardware experiments with a full multicopter model, investigate stronger theoretical guarantees for the adaptive online controller, compare against additional event-trigger formulations, and evaluate the approach under more challenging perception conditions including intermittent observations, latency, and occlusions.

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